

Subramanian Raj Narayanan

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EDUCATION

Master of Science in Data Science - Arizona State University

May 2026

Bachelor of Technology in Computer Science Engineering - SRM Institute of Science and Technology

May 2024

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, R

AI/ML & Gen AI: Transformers, LLMs, PyTorch, TensorFlow, Scikit-learn, Fine-tuning, LoRA/QLoRA, RAG Pipelines, Hybrid Retrieval, Reranking, NLP, Reinforcement Learning, RLHF, Prompt Engineering, Embeddings, Hugging Face, LlamaIndex, Computer Vision, Deep Learning, CUDA, spaCY

Agentic AI: LangChain, LangGraph, MCP, Multi-agent Orchestration, Tool Calling, Function Calling, AI Agents, Persistent Memory

MLOps & Cloud: Git, Docker, Kubernetes, Linux, FastAPI, CI/CD, MLflow, Weights & Biases, vLLM, Model Monitoring, AWS (SageMaker, Redshift),

Databases: PostgreSQL, MySQL, MongoDB, Neo4j, ChromaDB, Milvus, Pinecone, Weaviate, pgvector, FAISS, Vector Search, Semantic Search

Data Engineering: PySpark, Kafka, Airflow, Spark, ETL Pipelines, Data Pipelines, Databricks, Snowflake, Grafana

EXPERIENCE

AI/ML Engineer | Botco.ai | Scottsdale, Arizona

June 2026 – Present

- Fine tuned a **RoBERTa base transformer** into **5 class adverse event classifier** for **Sanofi's Beyfortus** program, landing on **macro F1 0.7594** as the best of **6 training rounds**, validating at **36 of 40** real world trigger phrases before deployment through **Sutter Health**.
- Solved a raw unlabeled **VAERS** dataset problem with a **dual verification pipeline** combining keyword rules and model agreement, reaching **91.7% labeling accuracy**, then hand fixing **358 ambiguous rows** and a contaminated class mismatched with antidepressant text.
- Trained a **GPT2 style autoregressive transformer** from scratch on **26M+ clinical events** across **4,733 residents**, building a first of its kind **Generative Clinical AI system** that achieved **AUROC 0.693** for fall, **0.637** for hospitalization, and **0.632** for mortality prediction.
- Leading **Phase 3 AL to Memory Care transition prediction** for **634 confirmed cases** across **6, 12, and 18 month** horizons, and developing a **CTCAE and PRO CTCAE adverse event classification pipeline** mapping clinical notes to **NCI toxicity grading** using **NLP normalization**.
- Reported** directly to the **CTO**, working hands on with **client stakeholders** to shape **production grade Clinical AI systems** across **LangChain, LangGraph, and Kubernetes** microservices, delivering **HIPAA compliant** solutions with full **MLOps observability**.

AI/ML Engineer Intern | Botco.ai | Scottsdale, Arizona

January 2026 – May 2026

- Reproduced the **ARES/ETHOS hospital mortality inference pipeline** on an **NVIDIA H100 GPU**, generating **20,000 future patient trajectories** via **20 Monte Carlo simulations** per example, achieving **AUROC 0.8561**, **AUPRC 0.2754**, and **Brier score 0.0158**.
- Configured the **EHRSHOT clinical ML benchmark** with **FEMR pipeline** and pretrained **CLMBR-T-base** weights from **Hugging Face** across **15 few-shot prediction tasks**, debugging **JAX, CUDA, cuDNN, and NumPy** conflicts blocking **GPU-based representation generation**.
- Engineered a scalable **EHR tokenization pipeline** normalizing **10,144 fragmented ALIS diagnosis codes** via **NLP standardization**, transforming heterogeneous longitudinal records into **ML-ready patient timelines**.
- Diagnosed **5 root causes** of near-random **Phase 1 AUC (0.50)** including **diagnosis code fragmentation, equal loss weighting, and context window mismatch**, driving **Phase 2 architectural fixes** that improved **AUROC by 19%**

Software Engineer Intern | Techavidity Business Solutions | Frisco, Texas

June 2025 – August 2025

- Built an **AI-driven health intelligence assistant** using **RAG pipelines** integrating **Neo4j knowledge graphs, ChromaDB, and GPT-based retrieval**, enabling **multi-hop medical reasoning** and scalable **semantic search**.
- Developed an autonomous mobile testing system using **multi-agent LLM architecture** (Planner, Executor, Verifier, Supervisor) with **computer vision**, achieving **95%+ task execution accuracy** and reducing manual UI testing effort.
- Designed an **AI-powered IaC security analyzer** leveraging **CodeBERT** and rule-based validation to detect **Terraform misconfigurations** and enforce **cloud security best practices**.
- Engineered a **cyber threat detection system** using **LSTM and BERT** models to identify **C2 communication patterns** in **DNS/HTTPS traffic**.

Research Intern | Centre of Electric Mobility SRMIST | Chennai, India

February 2023 – February 2024

- Built **Husky**, a campus **EV transport application** supporting **5,000+ students**, optimizing ride scheduling and routing across user groups.
- Developed a **Python-based routing engine** using **graph algorithms and priority scheduling** to dynamically adjust routes based on time, demand, and traffic patterns.
- Designed a **real-time prioritization model** with **simulated user demand**, rebalancing routes based on contextual needs while ensuring **fair access for differently abled users**.

PROJECTS

FocusMate – AI Co-Pilot for ADHD User

November 2025

- Built a **mobile AI productivity assistant** using **React Native and FastAPI**, integrating **Gmail and Google Calendar APIs** with **LLM-powered email triage, action-item extraction, and real-time voice input** for automated task management.

AskNeo – AI Query Engine with Knowledge Graphs and Semantic Search

June 2025

- Engineered a **hybrid RAG system** integrating **Neo4j (3K+ nodes, 12K+ edges)** and **ChromaDB** via **LangChain** to enable **multi-hop medical reasoning** across structured and semantic data.